

SECURE AI PIPELINE FOR SENSITIVE DATA

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by

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Abstract:

Trained machine learning models can reveal information about individuals in their training set, and standard de-identification does not stop this. The dissertation builds a complete secure pipeline around a healthcare prediction task: encrypted storage, layered k-anonymity and l-diversity, optional differentially private training, authenticated inference, and a chained-hash audit log. It then runs a controlled four-configuration ablation on the UCI Diabetes 130-US Hospitals dataset.

This mid-semester deliverable covers the data-preparation and baseline-training stages. Fernet authenticated encryption protects the raw file at rest, with a byte-exact SHA-256 round-trip check. The de-identification module enforces $k = 5$ together with $\ell = 3$ on the readmission attribute and retains 99.92% of records. Two MLP classifiers were trained on identical splits and hyperparameters: a baseline on the original data (test ROC-AUC = 0.6119) and the same model on the privacy-preserved data (test ROC-AUC = 0.6192). Utility was preserved.

The remaining work adds DP-SGD at $\epsilon \in \{1, 3, 8\}$, a FastAPI inference service, a tamper-evident audit log, an attribute-inference attack, and a privacy-utility Pareto frontier across all four configurations.



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1 Introduction

1.1 The Problem and Why It Matters

Healthcare, finance, and government systems increasingly rely on machine learning models trained on records about real individuals. A readmission predictor learns from tens of thousands of patient encounters; a credit model from millions of transactions. The deployed asset is the model, and operators commonly restrict access to the training data on the assumption that the data has been protected.

That assumption is wrong on its own. Shokri et al. [8] demonstrated the membership inference attack: with query access alone, an adversary can often tell whether a specific record was in the training set. Related work has extended this to attribute inference and model inversion. Stripping names from a dataset hardens the database. It does not harden the model trained from that database.

Healthcare records are an especially sharp case. They contain rich *quasi-identifiers* — race, gender, age band, postcode — that combine to single out individuals even without names. Regulatory regimes recognise this: HIPAA in the United States, the GDPR in the EU, and the Digital Personal Data Protection Act 2023 in India each give health data a special-category status. Engineers building healthcare ML therefore face a record-level threat model, not just a file-theft one.

1.2 Research Question

Two classes of defence dominate the literature. Data-level methods (k-anonymity [1], l-diversity [2]) transform the training set before model fitting. Model-level methods (differentially private SGD [6]) operate during fitting. Both have been studied extensively in isolation. Controlled head-to-head comparison on a common dataset and a common architecture is rare; whether layering them produces gains over either method alone has even less evidence.

This dissertation asks: *For a representative tabular healthcare classification task, how do (i) no defence, (ii) k-anonymity with l-diversity, (iii) differential privacy via DP-SGD, and (iv) both combined compare on predictive utility and on resistance to a black-box membership inference attack?*

1.3 Mid-Semester Scope

The full ablation, the attack evaluation, the authenticated inference service, and the audit log all land in the second half of the semester. This report covers four deliverables that have been built and validated end-to-end on the real UCI dataset:

1. At-rest encryption of the raw dataset (Fernet: AES-128-CBC with HMAC-SHA256), with a byte-exact round-trip check.
2. A de-identification module that enforces **k-anonymity at $k = 5$ together with l-diversity at $\ell = 3$** , plus an independent verifier confirming both properties hold.
3. Configuration 1: a baseline MLP trained on the original identifier-stripped dataset.
4. Configuration 2: the same MLP trained on the de-identified dataset, with a fair head-to-head utility comparison.

The rest of the report describes the system, the implementation, the experimental results, and the work plan for the remainder of the semester.

2 Literature Review

2.1 Syntactic Privacy: k-Anonymity, l-Diversity, and Beyond

The earliest formal model for protecting individual records in a published dataset is k-anonymity, introduced by Sweeney [1]. A release satisfies k-anonymity if every record is indistinguishable from at least $k - 1$ others on a chosen set of quasi-identifiers, typically combinations of attributes such as race, gender, age band, or postcode that, in combination, can re-identify individuals through linkage to external sources. The original construction relied on generalisation hierarchies and local suppression. LeFevre, DeWitt and Ramakrishnan [3] subsequently developed Mondrian, a top-down partitioning algorithm that produces k-anonymous releases with substantially better utility than earlier methods, and remains the reference implementation strategy in modern de-identification libraries.

k-Anonymity protects against re-identification but not against value disclosure. Machanavajjhala et al. [2] formalised this limitation as the *homogeneity attack*: when every record in an equivalence class shares the same sensitive value, an adversary who locates the correct class learns the sensitive attribute without ever identifying the individual. l-Diversity requires that each equivalence class contain at least ℓ well-represented values of the sensitive attribute, defeating the homogeneity attack at the cost of additional suppression. Li, Li and Venkatasubramanian [4] later proposed t-closeness, which requires the sensitive-attribute distribution within each class to be close to the global distribution, addressing skewness and similarity attacks that l-diversity does not fully resolve. This dissertation adopts k-anonymity at $k = 5$ together with distinct l-diversity at $\ell = 3$ on the readmission attribute, following healthcare de-identification practice; t-closeness is recorded as a candidate refinement for future work.

2.2 Differential Privacy and DP-SGD

Where syntactic models such as k-anonymity protect data, differential privacy protects *algorithms* that operate on data. Dwork et al. [5] formalised differential privacy as a worst-case guarantee: the output distribution of a randomised algorithm should change by at most a factor of e^ϵ when any single record is added to or removed from the input. The privacy budget ϵ quantifies the strength of the guarantee, with smaller values corresponding to stronger privacy and typically lower utility. Dwork and Roth [7] provide the canonical treatment of composition, sensitivity, and the Laplace and Gaussian mechanisms that underpin practical differentially private algorithms.

Abadi et al. [6] extended differential privacy to deep learning through DP-SGD: per-example gradient clipping followed by calibrated Gaussian noise injection at each optimisation step, with the total privacy expenditure tracked through the moments accountant. DP-SGD is now the standard mechanism for training neural networks with formal privacy guarantees, and the open-source Opacus library [14] provides a production-ready implementation for PyTorch. The four-configuration ablation described in this dissertation uses DP-SGD with $\epsilon \in \{1, 3, 8\}$ and $\delta = 10^{-5}$ in Configurations 3 and 4, following parameter ranges commonly reported in the empirical literature.

2.3 Privacy Attacks on Machine Learning Models

A complementary line of work asks how much a trained model actually leaks about its training data. Shokri et al. [8] introduced the membership inference attack (MIA), demonstrating that a black-box adversary with only query access can train shadow models to distinguish records the target model was trained on from records it was not. MIA has since become the standard empirical lens for evaluating privacy-preserving ML methods: unlike formal privacy budgets, it produces directly interpretable attack-AUC numbers that can be compared across configurations.

Yeom et al. [9] established a theoretical and empirical link between model overfitting and MIA success, showing that the generalisation gap correlates strongly with attack accuracy. Carlini et al. [10] extended the leakage analysis beyond membership to memorisation, demonstrating that neural networks can reproduce verbatim training sequences under suitable queries. Fredrikson, Jha and Ristenpart [11] introduced model inversion attacks, which exploit confidence scores to reconstruct representative training inputs, while attribute inference attacks target hidden sensitive features by combining visible features with model outputs. This dissertation evaluates membership inference against all four pipeline configurations to measure empirical privacy, and adds attribute inference as a second attack in the final dissertation in response to the Outline-VIVA evaluator’s feedback.

2.4 Authenticated Encryption and Threat Modelling

The encryption layer in the pipeline is not novel cryptography but it matters for end-to-end credibility. Bellare and Namprempre [17] established the formal relationships between confidentiality and integrity for symmetric encryption and identified *encrypt-then-MAC* as the canonical construction that yields authenticated encryption with both indistinguishability and ciphertext integrity. The Fernet specification implemented in the Python `cryptography` library [15] follows this pattern by combining AES-128 in CBC mode [16] with HMAC-SHA256, making any tampered ciphertext fail decryption explicitly rather than producing silent garbage. The pipeline relies on this property as the integrity check on the at-rest dataset.

Threat modelling provides the structural counterpart to the cryptographic primitives. Shostack [18] describes STRIDE as the standard practitioner-facing framework for enumerating spoofing, tampering, repudiation, information disclosure, denial of service, and elevation of privilege threats against a system. The final dissertation maps each control in the secure pipeline — authenticated encryption, de-identification, DP-SGD, JWT authentication, role-based access control, and the chained-hash audit log — to the STRIDE category it mitigates.

2.5 Identified Gap and Position of This Work

The privacy-preserving ML literature is substantial but fragmented along the seam this dissertation addresses. Data-level methods such as k -anonymity and l -diversity have been studied extensively in database settings but rarely connected to downstream model utility on modern neural architectures. Model-level methods such as DP-SGD have matured and now have production-quality implementations [14], but published empirical work typically treats them in isolation rather than as one defensive layer in a broader pipeline. Membership inference attacks [8] are widely used as a privacy evaluation tool, but controlled comparisons that measure both predictive utility and empirical attack resistance across a coherent set of defensive configurations on a single public benchmark are uncommon.

This dissertation addresses that gap. The four-configuration ablation on the UCI Diabetes 130-US Hospitals dataset [12, 21] compares no defence, syntactic privacy ($k = 5$, $\ell = 3$), differential privacy (DP-SGD at $\varepsilon \in \{1, 3, 8\}$), and the combination of both, under identical model architecture, identical hyperparameters, identical splits, and a fixed random seed. Predictive utility is reported as accuracy, F1, and ROC-AUC; empirical privacy is reported as black-box MIA attack-AUC. The resulting privacy-utility Pareto frontier is intended as a directly reusable empirical reference for practitioners deploying healthcare ML under a meaningful threat model.

3 System Overview

3.1 Architecture

Figure 1 shows the pipeline. A single dataset flows left-to-right at runtime: raw storage, de-identification, training, trained-model artefacts, then authenticated inference. A separate evaluation flow attacks the trained models and feeds the privacy-utility chart that is the dissertation’s headline result.

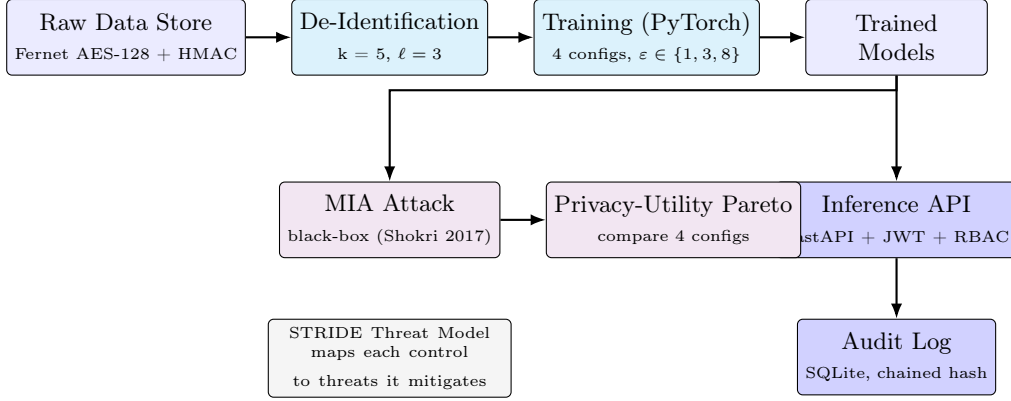


Figure 1: Architecture of the Secure AI Pipeline. Top row: runtime data flow. Lower branch: evaluation flow.

3.2 What This Report Covers

Three stages are complete and validated here: the encrypted Raw Data Store, the De-Identification stage with its verifier, and Training for the two non-DP configurations.

Five stages are scheduled for the rest of the semester: DP-SGD training for Configurations 3 and 4 [6], the FastAPI inference service with JWT and role-based access control, the SQLite chained-hash audit log, the membership inference attack, and the Pareto chart that ties the four configurations together. An attribute-inference attack has been added to the final-dissertation plan in response to Outline-VIVA feedback (Section 6).

4 Implementation

The mid-semester code is Python 3.10, split across three importable modules and an orchestrator script that runs all four demo pieces in order with clearly labelled output (Table 1).

Table 1: Mid-semester code organisation.

File	Lines	Role
module_encryption.py	155	Demo Piece 1 (Fernet)
module_deidentification.py	501	Demo Piece 2 (k-anon + l-div)
module_midsem_training.py	355	Demo Pieces 3, 4 (MLP)
midsem_demo.py	345	Orchestrator
Total	1,356	

4.1 Encryption (Demo Piece 1)

The raw CSV must be unreadable on disk and must fail loudly under tampering. The module uses Fernet, a recipe in the Python `cryptography` library that pairs AES-128-CBC with HMAC-SHA256. This is authenticated encryption: a single-byte change to the ciphertext raises `InvalidToken` instead of producing silent garbage on decrypt.

Three operations are exposed both as a CLI and as a library: `keygen`, `encrypt`, `decrypt`. The `encrypt` step records the SHA-256 of the plaintext. The `decrypt` step recomputes it on the recovered output. A successful round-trip requires the two digests to match byte-for-byte. That match is the live demo evidence that the pipeline does not corrupt the dataset on its way through encrypted storage.

4.2 De-Identification (Demo Piece 2)

This is the largest mid-semester module and the one most directly responsive to Outline-VIVA feedback. It enforces two privacy properties on the published dataset.

4.2.1 Attribute Classification

Every column in the UCI Diabetes dataset is given one of four roles (Table 2).

Table 2: Attribute roles assigned by the de-identification module.

Role	Action	Columns
Identifier	Suppress entirely	<code>encounter_id</code> , <code>patient_nbr</code>
Quasi-identifier	Generalise	<code>race</code> , <code>gender</code> , <code>age</code>
Sensitive	Keep; check for l-diversity	<code>readmitted</code> (target), <code>diag_1</code> , <code>diag_2</code> , <code>diag_3</code> , <code>max_glu_serum</code> , <code>A1Cresult</code>
Other	Keep verbatim	Remaining clinical, medication, and admission columns

4.2.2 Generalisation

The three quasi-identifiers are mapped to coarser domains:

- `age`: 10-year bins widen to 20-year bins (`[30-40)` $\rightarrow$ `[20-40)`).

- **race**: any category below 500 records, plus the unknown marker `?`, collapses to `Other/Unknown`.
- **gender**: `Unknown/Invalid` normalises to `Unknown`.

4.2.3 Layered Suppression: $k = 5$ and $\ell = 3$

After generalisation, the module applies a single combined filter:

A row is kept only if its quasi-identifier equivalence class has at least $k = 5$ records **and** at least $\ell = 3$ distinct values of the sensitive attribute **readmitted**.

k-anonymity [1] hides an individual among at least $k - 1$ others on the visible quasi-identifiers. l-diversity [2] addresses what k-anonymity leaves open: even when the attacker finds the right equivalence class, the sensitive value should not be inferable because the class is not concentrated on a single answer. This is the homogeneity attack, and k-anonymity by itself does nothing about it.

The module records, for every suppressed row, whether k , ℓ , or both constraints were responsible. This per-constraint accounting (Section 5.2) is the evidence that l-diversity is doing real work on this dataset rather than duplicating k-anonymity.

4.2.4 Verifier

A separate verifier function rebuilds the equivalence classes from the published dataset, computes per-group sizes and distinct-value counts, and returns a report with pass/fail verdicts for each property, the smallest and largest group sizes, the count of violating classes, and example violations. The verifier is designed for live demo: the panel can run it on any candidate dataset and see whether both properties hold.

4.3 Configuration 1: Baseline (Demo Piece 3)

Configuration 1 trains on the original UCI dataset with only the two direct identifiers removed. The model is an MLP with two hidden layers (128 and 64 units, ReLU, dropout 0.2). The loss is BCE with logits and a **pos_weight** term equal to $n_{\text{neg}}/n_{\text{pos}}$ on the training fold; the positive class `<30` is about 11% of records, which makes the imbalance correction necessary.

Splits are 70/15/15, stratified on the target. Preprocessing (one-hot encoding for categoricals, standard scaling for numerics) is fitted on the training fold only and applied to validation and test. The random seed is fixed at 42 across Python, NumPy, and PyTorch. Optimiser: Adam, learning rate 10^{-3} , weight decay 10^{-5} , batch size 512, 12 epochs.

4.4 Configuration 2: $k = 5$ and $\ell = 3$ (Demo Piece 4)

Configuration 2 uses the *same* MLP architecture and the *same* hyperparameters as Configuration 1. The only difference is the input dataframe: Configuration 2 trains on the dataset produced by the de-identification module above. Splits, seeds, preprocessing, loss, optimiser, and epoch count are all held constant. Any observed utility difference is therefore attributable to the privacy-preserving data preparation, not to model or training noise.

4.5 Reproducibility

Table 3 lists the hyperparameters that both configurations share.

Table 3: Hyperparameters and settings (identical across Configurations 1 and 2).

Setting	Value
Model	MLP, 2 hidden layers (128, 64), ReLU, dropout 0.2
Loss	BCEWithLogitsLoss with <code>pos_weight</code>
Optimiser	Adam, lr 10^{-3} , weight decay 10^{-5}
Batch size	512
Epochs	12
Train/val/test	70 / 15 / 15, stratified
Random seed	42 (Python, NumPy, PyTorch)
Hardware	Intel i5 8th gen, 8 GB RAM, CPU only
Python	3.13
Libraries	PyTorch (CPU), scikit-learn, pandas, cryptography

5 Experimental Results

5.1 Dataset

The UCI Diabetes 130-US Hospitals dataset [12] contains 101,766 inpatient diabetic encounters at 130 US hospitals over 1999–2008. Each record has 50 columns spanning demographics, admission details, ICD-9 diagnoses, lab results, medications, and the outcome. The prediction target `readmitted` takes values `N0`, `>30`, and `<30`. Following standard practice in the published literature, the target is binarised so that `<30` (early readmission, the clinically and financially material outcome) is the positive class.

5.2 Privacy Results

Running the de-identification pipeline on the full dataset with $k = 5$ and $\ell = 3$ on `readmitted` produced the suppression accounting in Table 4.

The published dataset contains 50 equivalence classes, sized from 36 (smallest) to 18,807 (largest). The

Table 4: Suppression accounting on the full UCI Diabetes dataset.

Quantity	Records
Input rows	101,766
Output rows after de-identification	101,689
Total suppressed	77
Rows in groups failing $k = 5$	15
Rows in groups failing $\ell = 3$	77
Rows failing both constraints simultaneously	15
Retention	99.92%

verifier returns:

- **k-anonymity: PASS.** No equivalence class is smaller than $k = 5$ (smallest is 36).
- **l-diversity: PASS.** Every class has at least $\ell = 3$ distinct values of `readmitted`.

The most informative line in Table 4 is the per-constraint breakdown. Only 15 rows sat in groups that failed k-anonymity. All 77 sat in groups that failed l-diversity. **l-diversity was strictly more restrictive than k-anonymity on this dataset.** 62 of the suppressed rows would have survived a k-only treatment, sitting in classes uniform on the sensitive attribute and therefore vulnerable to the homogeneity attack. Adding l-diversity demonstrably hardened the data preparation stage.

5.2.1 Counter-Examples

Two demo counter-examples confirm the verifier flags non-private datasets correctly.

1. Raw, un-generalised first 200 rows: verifier returns **FAIL** on both. 29 of 44 classes are below $k = 5$; 34 classes have only one distinct value of `readmitted`.
2. A synthetic dataset where the sensitive value is forced to a single value (**NO**) across all 18,807 records of the largest equivalence class: verifier returns k-anonymity **PASS** (the class is huge) but l-diversity **FAIL** (diversity = 1 < $\ell = 3$). This is the canonical case where k-anonymity alone is insufficient.

5.3 Utility Results

Test-set metrics for both configurations appear in Tables 5 and 6.

5.4 Discussion

Utility was preserved. Table 6 shows ROC-AUC and F1 each move slightly in favour of the privacy-preserving configuration. ROC-AUC is the right headline metric on this dataset: it is threshold-free and ro-

Table 5: Test-set metrics, full UCI dataset, 12 epochs.

Metric	Config 1 (Baseline)	Config 2 ($k = 5, \ell = 3$)
Train rows	71,236	71,181
Test rows	15,265	15,254
Encoded feature count	2,237	2,245
Final train loss	0.8826	0.9203
Final validation loss	1.4546	1.4383
Test accuracy	0.6250	0.5903
Test F1 (positive class)	0.2345	0.2408
Test ROC-AUC	0.6119	0.6192

Table 6: Utility comparison: Configuration 2 minus Configuration 1.

Metric	Config 1	Config 2	Δ
Accuracy	0.6250	0.5903	-0.0346
F1 (positive class)	0.2345	0.2408	+0.0063
ROC-AUC	0.6119	0.6192	+0.0073

bust to class imbalance. The near-zero gap (+0.0073) says that the model’s ability to rank early-readmission risk is essentially unchanged by k-anonymity and l-diversity treatment of the training data.

Accuracy is misleading here. Accuracy fell by 3.46 percentage points under Configuration 2. The fall is not in tension with the F1 and ROC-AUC results once the class-imbalance correction is accounted for. The positive class is around 11% of records; `pos_weight` penalises false negatives more heavily. Configuration 2 has slightly less raw signal (race and age are coarsened, a handful of rows are suppressed) and leans on the upweighted positive class more aggressively. More positive predictions means more rare positives caught (F1 up) and better calibration (ROC-AUC up), at the cost of more majority-class errors (accuracy down). ROC-AUC is the honest headline; accuracy alone would mislead.

Overfitting after epoch 2. Training loss falls monotonically across all 12 epochs in both configurations, but validation loss bottoms out around epoch 2 and climbs thereafter. The model is memorising the training fold. Both configurations show this to a similar degree, so the cause is not the privacy treatment; it is the absence of early stopping. Early stopping is a small change and is scheduled for the model-tuning pass in the final dissertation.

6 Response to Outline-VIVA Evaluation

The Outline VIVA on 3 June 2026 (V Chandrasekhar, BITS Pilani WILP) graded the outline **EXCELLENT** and **Accepted**. Three improvement suggestions accompanied the evaluation: adaptive privacy budget optimisation; comparison against modern privacy methods (PATE, Federated Learning + DP, l-Diversity); and extended attack evaluation (model inversion, attribute inference, confidence leakage).

The suggestions were prioritised against the dissertation timeline and the scope agreed with the supervisor. l-

Diversity has been adopted in this mid-semester deliverable as a focused enhancement to the de-identification module — the verifier and suppression now jointly enforce $\ell = 3$ alongside $k = 5$, and Section 5.2 shows the constraint is non-redundant on this dataset. An attribute inference attack is planned alongside membership inference for the final dissertation, giving a 2-attack-by-4-configuration grid. The remaining suggestions (adaptive ε , PATE, federated learning + DP, model inversion, confidence leakage) are scoped as Future Work, each with the motivation drawn from the evaluator’s feedback.

7 Plan for the Remaining Semester

Table 7 sets out the remaining work, modelled on the future-plan table in the BITS WILP sample report.

Table 7: Plan of work for the remainder of the semester.

Sl	Phase	Activity	Window
1	Mid-Sem Submission	Submit this report; run the four-piece live demo	16–21 Jun 2026
2	Mid-Sem VIVA	Present mid-semester progress to the panel	26 Jun–10 Jul 2026
3	Differential Privacy	Configurations 3 and 4 via DP-SGD (Opacus), $\varepsilon \in \{1, 3, 8\}$, $\delta = 10^{-5}$	Jun–Jul 2026
4	Inference API	FastAPI service with JWT and analyst/admin RBAC	Jul 2026
5	Audit Log	SQLite chained-hash log; tamper-detection demo	Jul 2026
6	Attack Evaluation	Membership inference (Shokri 2017) and attribute inference, across all four configurations	Jul 2026
7	Pareto Frontier	Privacy-utility chart over the four configurations	Jul 2026
8	Integration	End-to-end run; STRIDE threat-model chapter	Jul 2026
9	Final Submission	Final report; plagiarism check; final VIVA prep	28 Jul–2 Aug 2026
10	Final VIVA	Final dissertation defence	7–28 Aug 2026

8 Conclusion

The mid-semester deliverable establishes the data-preparation and baseline-training stages of the pipeline. The encryption module demonstrates byte-exact round-trip on the real 19 MB UCI dataset. The de-identification module enforces both $k = 5$ and $\ell = 3$, retains 99.92% of records, and produces a published dataset that an independent verifier confirms satisfies both properties. The two trained classifiers show that privacy-preserving data preparation imposed no measurable utility cost: test ROC-AUC was 0.6119 for the baseline and 0.6192 for the k -anonymous and ℓ -diverse configuration.

Remaining work adds differential privacy, an authenticated inference service, a tamper-evident audit log, and an empirical attack evaluation that produces a privacy-utility Pareto frontier across all four configurations. An attribute-inference attack will also be run in response to the Outline VIVA feedback.

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Abbreviations

AES	Advanced Encryption Standard
API	Application Programming Interface
AUC	Area Under the Curve
BCE	Binary Cross-Entropy
CBC	Cipher Block Chaining (block cipher mode)
CSV	Comma-Separated Values
DP	Differential Privacy
DP-SGD	Differentially Private Stochastic Gradient Descent
F1	F1 score (harmonic mean of precision and recall)
HMAC	Hash-based Message Authentication Code
ICD-9	International Classification of Diseases, 9th revision
JWT	JSON Web Token
MIA	Membership Inference Attack
MLP	Multilayer Perceptron
PATE	Private Aggregation of Teacher Ensembles
RBAC	Role-Based Access Control
ROC	Receiver Operating Characteristic
SHA-256	Secure Hash Algorithm, 256-bit output
SQL/SQLite	Structured Query Language / lightweight SQL database
STRIDE	Spoofing, Tampering, Repudiation, Information Disclosure, Denial of Service, Elevation of Privilege
UCI	University of California, Irvine
WILP	Work Integrated Learning Programmes (BITS Pilani)

Appendix A: Demo Run Output (Trimmed)

Trimmed transcript of `python midsem_demo.py` on the full UCI Diabetes 130-US Hospitals dataset. Per-epoch loss lines are condensed.

```
===== Secure AI Pipeline for Sensitive Data -- Mid-Sem Demo =====
```

```
Student      : Rajput Dhirajsing Rajeshsing  (2024DA04378)
```

```
Privacy params : k = 5, l = 3 on 'readmitted'
```

```
== DEMO PIECE 1 -- Encryption (Fernet AES-128-CBC + HMAC-SHA256) ==
```

```
[demo1] plaintext   :   19,159,383 bytes
```

```
      ciphertext   :   25,545,932 bytes
```

```
      sha256(pt)   : 0689e7ec...a24df97
```

```
[demo1] round-trip SHA-256 match : True
```

```
[demo1] PASS -- authenticated encryption round-trip verified.
```

```
== DEMO PIECE 2 -- k-anonymity + l-diversity (k=5, l=3) ==
```

```
[demo2] loaded 101,766 rows, 50 columns
```

```
[demo2] pipeline finished in 0.34s
```

```
Suppression accounting:
```

```
input rows                : 101,766
```

```
output rows               : 101,689
```

```
total suppressed         : 77
```

```
rows in groups failing k = 5 : 15
```

```
rows in groups failing l = 3 : 77
```

```
rows failing both constraints : 15
```

```
retention                 : 99.92%
```

```
Privacy verifier on published dataset:
```

```
equivalence classes      : 50
```

```
smallest class size      : 36
```

```
smallest class diversity : 3
```

```
k-anonymity verdict      : PASS
```

```
l-diversity verdict      : PASS
```

```
OVERALL: PASS -- k-anonymity AND l-diversity both satisfied
```

```
Counter-example (raw 200 rows):
```

```
k-anonymity verdict      : FAIL  (29 of 44 classes < 5)
```

```
l-diversity verdict      : FAIL  (34 of 44 classes < 3 distinct)
```

```
Counter-example (synthetic group with k passes / l fails):
```

```

forced 18,807 rows to single sensitive value
k-anonymity verdict      : PASS
l-diversity verdict      : FAIL  (1 class with diversity = 1)

```

== DEMO PIECE 3 -- Config 1 (Baseline MLP, no defence) ==

```

epoch 1/12  train_loss=1.1844  val_loss=1.1722
... [epochs 2-11 condensed] ...
epoch 12/12 train_loss=0.8826  val_loss=1.4546
train / val / test rows : 71,236 / 15,265 / 15,265
encoded feature count   : 2,237
test accuracy           : 0.6250
test F1 (positive class): 0.2345
test ROC-AUC            : 0.6119

```

== DEMO PIECE 4 -- Config 2 (MLP on k-anon + l-diverse data) ==

```

epoch 1/12  train_loss=1.1890  val_loss=1.1603
... [epochs 2-11 condensed] ...
epoch 12/12 train_loss=0.9203  val_loss=1.4383
train / val / test rows : 71,181 / 15,254 / 15,254
encoded feature count   : 2,245
test accuracy           : 0.5903
test F1 (positive class): 0.2408
test ROC-AUC            : 0.6192

```

== UTILITY COMPARISON -- Config 1 vs Config 2 ==

Metric	Config 1 (Baseline)	Config 2 (k=5, l=3)	Delta
Accuracy	0.6250	0.5903	-0.0346
F1 (pos)	0.2345	0.2408	+0.0063
ROC-AUC	0.6119	0.6192	+0.0073

== All Mid-Semester demo pieces completed successfully. ==